AMR Awareness & Risk Checker
Hackathon Project Report

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Abstract

Antimicrobial Resistance (AMR) is one of today's biggest public health crises. Because antibiotics are frequently misused—like taking them for viral infections or stopping doses early—bacteria are adapting, making standard treatments less effective. While doctors know the risks, much of the public remains unaware. To bridge this gap, we built the AMR Awareness & Risk Checker for the KBG Club Recruitment Hackathon at IIT Mandi. Rather than diagnosing illnesses, this educational web app teaches users about AMR through a guided risk assessment, an AI assistant, reliable reference materials, and a "Myth vs. Fact" module. By combining a rule-based engine with responsible AI, the app delivers safe, transparent guidance while always encouraging users to consult a doctor.

1. Problem Statement

Antimicrobial Resistance (AMR) is rapidly becoming one of the greatest threats to global healthcare. Because antibiotics are continuously misused, bacteria are evolving faster than we can develop new treatments. This directly threatens routine surgeries, cancer therapies, and basic infection management.

The root cause is a severe public knowledge gap. Individuals frequently self-medicate, stop treatments early, or use antibiotics for viral infections like colds and flu—driving up healthcare costs and treatment failures. Currently, there is a critical lack of accessible, interactive tools to educate the public on responsible antibiotic use. We need a solution that bridges this gap, dispels medical myths, and connects users to trusted health information from the WHO, CDC, and ICMR without acting as a diagnostic tool.

2. Objectives

To address the critical public health challenge of antimicrobial resistance, the project focuses on the following primary objectives:

- Increase public awareness regarding antimicrobial resistance.
- Encourage responsible antibiotic usage and discourage self-medication.
- Provide a transparent risk assessment through a structured questionnaire.
- Deliver educational responses using an AI assistant constrained by trusted public-health guidance.

- Correct common misconceptions through an evidence-based Myth vs Fact section.
- Promote consultation with qualified healthcare professionals rather than self-diagnosis.

3. Motivation and Background

Antimicrobial Resistance (AMR) is a major global issue driven by the excessive and incorrect use of antibiotics in healthcare, agriculture, and self-medication. Despite ongoing campaigns by health organizations, public awareness remains low. Many people still think antibiotics cure viral infections, stop taking them early when they feel better, or use leftover pills without consulting a doctor. These habits speed up the growth of resistant bacteria strains.

Because of this, healthcare systems globally are shifting their focus toward preventative education instead of just treating advanced, drug-resistant infections. Digital tools offer a powerful way to scale accurate information and change user behavior. We built this project to turn dense, complex public health guidelines into an interactive, user-friendly web app that educates people before antibiotic misuse happens.

4. Literature and Reference Study

Our application is built directly on public guidelines from the WHO, CDC, and ICMR, which stress public education and responsible antibiotic use as key to slowing down resistance. Right now, most educational tools are just static webpages or fact sheets that lack interactive engagement. On the flip side, unconstrained AI chatbots risk giving confident but dangerously misleading medical advice.
Our project bridges this gap. By combining strict rule-based decision logic with a highly constrained AI assistant, we ensure users get engaging, educational information grounded in trusted public-health data, without crossing into diagnostic advice.

5. System Overview

The AMR Awareness & Risk Checker uses a modular design focused on safety, maintenance, and user experience. When users open the app, they first see a mandatory disclaimer stating that the platform provides only educational information, not medical advice. Users then complete a questionnaire about their antibiotic habits, which a rule-based engine scores to categorize their awareness levels. If answers suggest a viral illness, the app redirects users to home-care tips instead of antibiotic info. Afterward, users can chat with an AI educational assistant or explore "Myth vs. Fact" section and reference guides built from WHO, CDC, and ICMR data.

5.1. Design Philosophy

Our primary design principle was "Education, Not Diagnosis." We built every module to minimize misinformation and maximize transparency. By combining rule-based scoring, verified references, safety disclaimers, and a tightly constrained AI, we ensure users receive trustworthy guidance that never encourages self-medication or replaces professional medical care.

6. System Architecture

The AMR Awareness & Risk Checker app is split into distinct functional layers: the user interface, application logic, educational AI assistant, configuration module, and a trusted knowledge layer. We used Streamlit to render an intuitive web interface, while Python handles the backend interactions between the questionnaire, scoring engine, chatbot, and reference database.

A key design choice is the strict separation between rule-based logic and AI-generated text. Risk scores are calculated using strictly predefined rules to ensure transparent, repeatable results. Meanwhile, the AI assistant only handles educational questions, kept in check by strict prompting and official reference text. This hybrid approach keeps conversations natural while ensuring the app never hands out unsafe medical advice.

7. Workflow of the Application

The application follows a straightforward, step-by-step workflow:

1. **Safety Disclaimer:** Users must first acknowledge a mandatory disclaimer explaining the app's educational focus.
2. **Risk Assessment:** Users complete a structured questionnaire about antibiotic use and health awareness. A rule-based scoring engine processes the answers to place users into predefined risk bands.
3. **Smart Redirection:** If the responses point toward a viral illness (where antibiotics won't help), the app routes the user to home-care guidance before showing educational recommendations.
4. **Interactive Learning:** Once finished, users can ask the AI assistant AMR-related questions, supported by official WHO, CDC, and ICMR snippets in the sidebar.
5. **Myth vs. Fact:** The workflow ends with an interactive game to clear up common misconceptions and reinforce good habits.

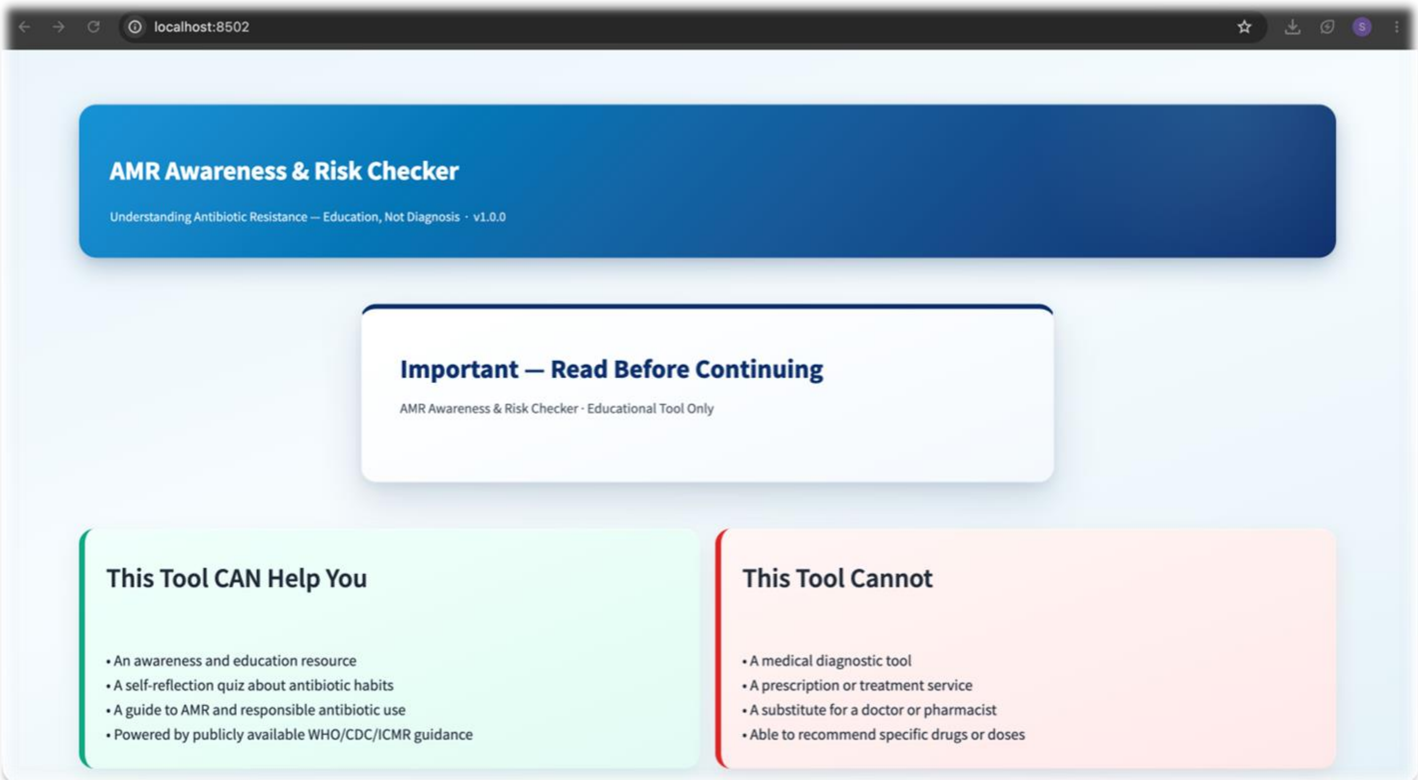


Fig1. Home Page of the AMR Awareness & Risk Checker

Our landing page introduces users to the AMR Awareness & Risk Checker and immediately sets clear expectations. It features a professional blue gradient banner displaying the project title, software version, and the core subtitle: "Understanding Antibiotic Resistance – Education, Not Diagnosis."

To ensure safety and ethical transparency, the homepage includes:

1. **Mandatory Safety Disclaimer:** A "Read Before Continuing" section clarifying that the platform is purely educational and does not replace professional medical advice.
2. **"This Tool CAN Help You" Panel:** Briefly explains how the app assesses antibiotic habits, promotes responsible usage, and provides insights built on WHO, CDC, and ICMR guidelines.
3. **"This Tool CANNOT" Panel:** Explicitly states that the app does not diagnose illnesses, prescribe medication, recommend dosages, or replace healthcare professionals.

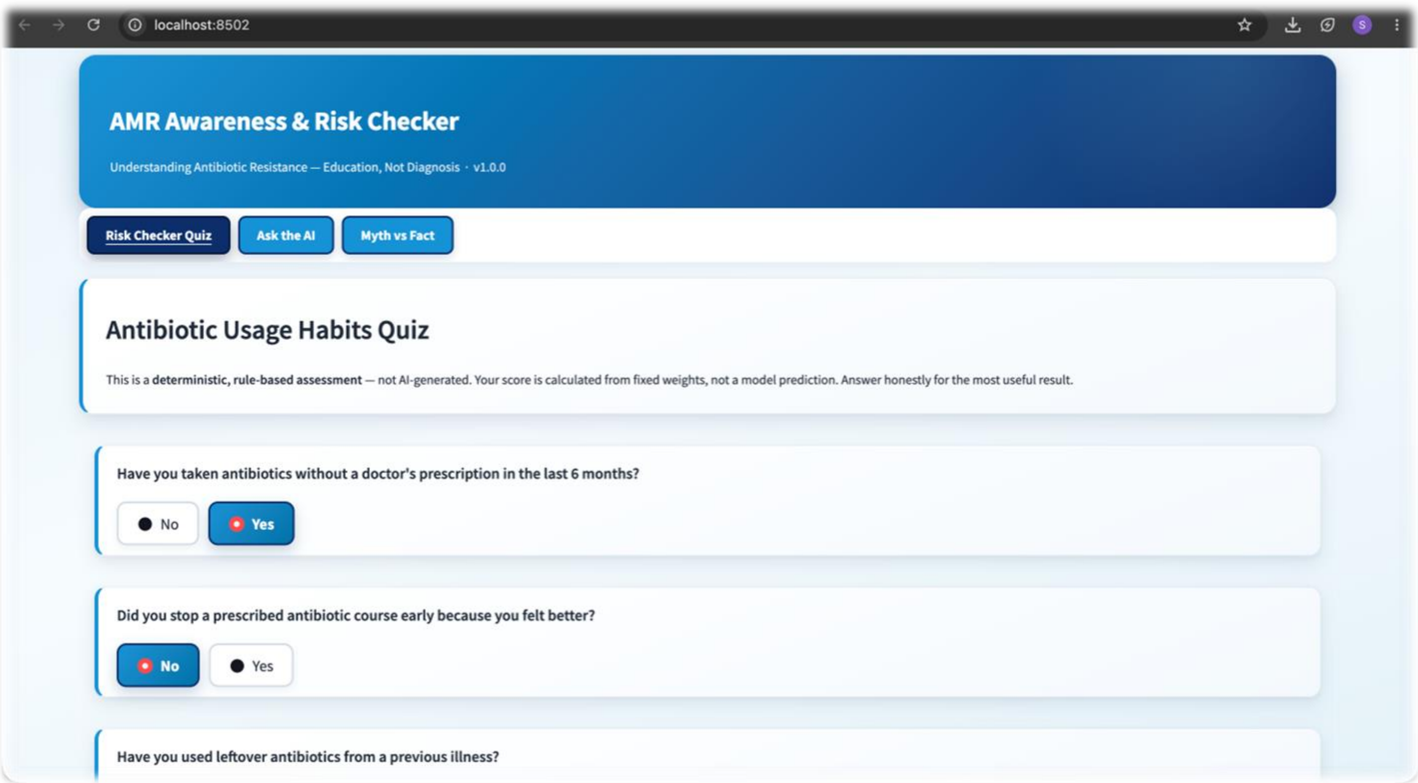


Fig2. Antibiotic Usage Habits Quiz

The second interface hosts the Risk Checker Quiz, the core assessment module of the application. The page includes navigation buttons so users can easily switch between the Quiz, "Ask the AI," and "Myth vs. Fact" modules while keeping a consistent layout.

Key features of this page include:

1. **Rule-Based Transparency:** A descriptive note clarifies that the assessment uses predefined, weighted rules rather than predictive machine learning or AI models, ensuring transparent risk calculation.
2. **Behavioral Questionnaire:** Users answer clear Yes/No questions targeting common habits like self-medication, stopping prescriptions early, and reusing leftover antibiotics.
3. **Personalized Feedback:** The scoring engine processes these answers to generate a risk score, which instantly unlocks tailored educational guidance and awareness messages.

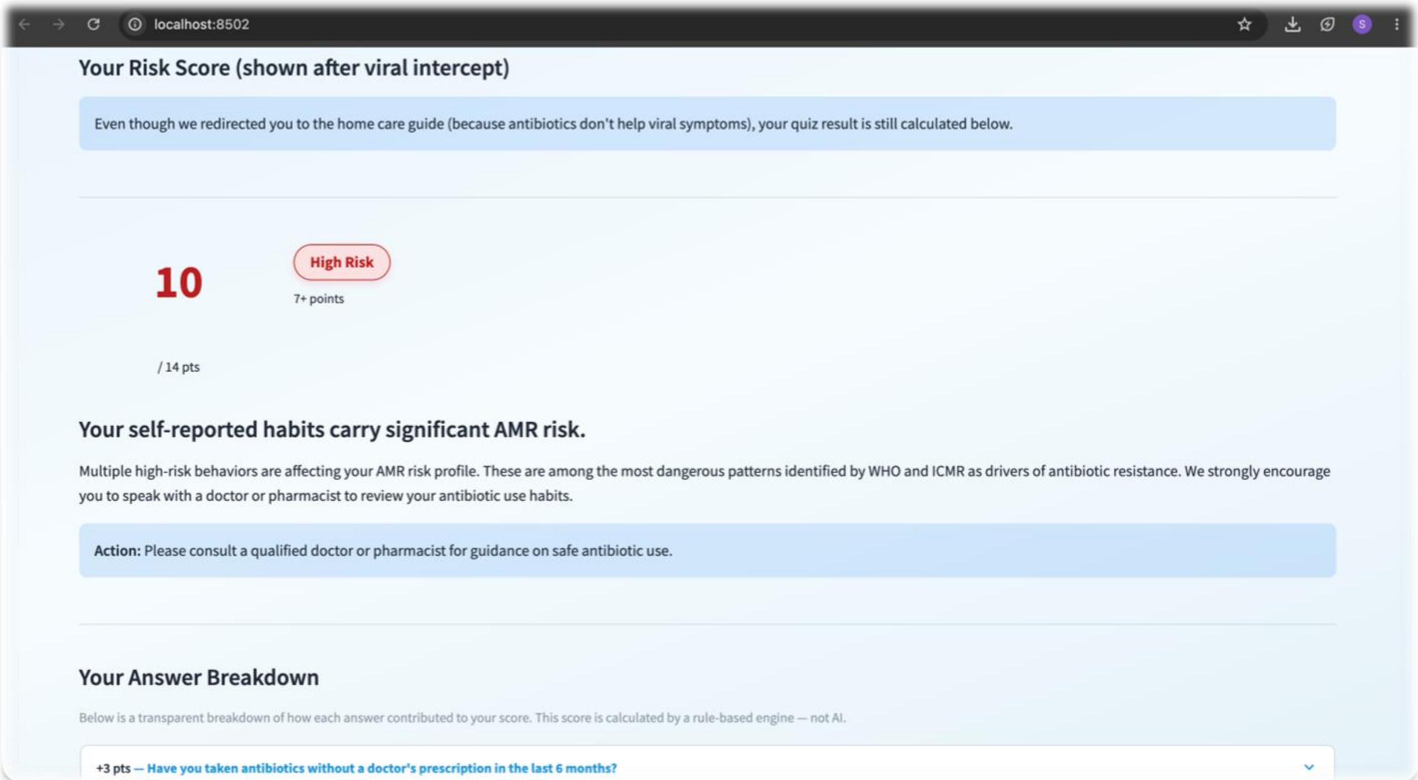


Fig3. AMR Risk Assessment Results

The third interface displays the Results page generated immediately after a user completes the Antibiotic Usage Habits Quiz.

Key elements of this webpage include:

1. **Transparent Risk Scoring:** A rule-based scoring engine calculates the final points to sort users into predefined risk bands (such as the "High Risk" category).
2. **Educational Impact:** The page explicitly shows how specific behaviors drive up antimicrobial resistance for both the individual and the community, steering completely clear of clinical diagnosis.
3. **Advisory & Breakdown Panels:** A prominent advisory banner reminds users to consult a healthcare professional. Below it, an Answer Breakdown section lets users see exactly which habits inflated their risk score, driving the behavioral changes needed to combat AMR.

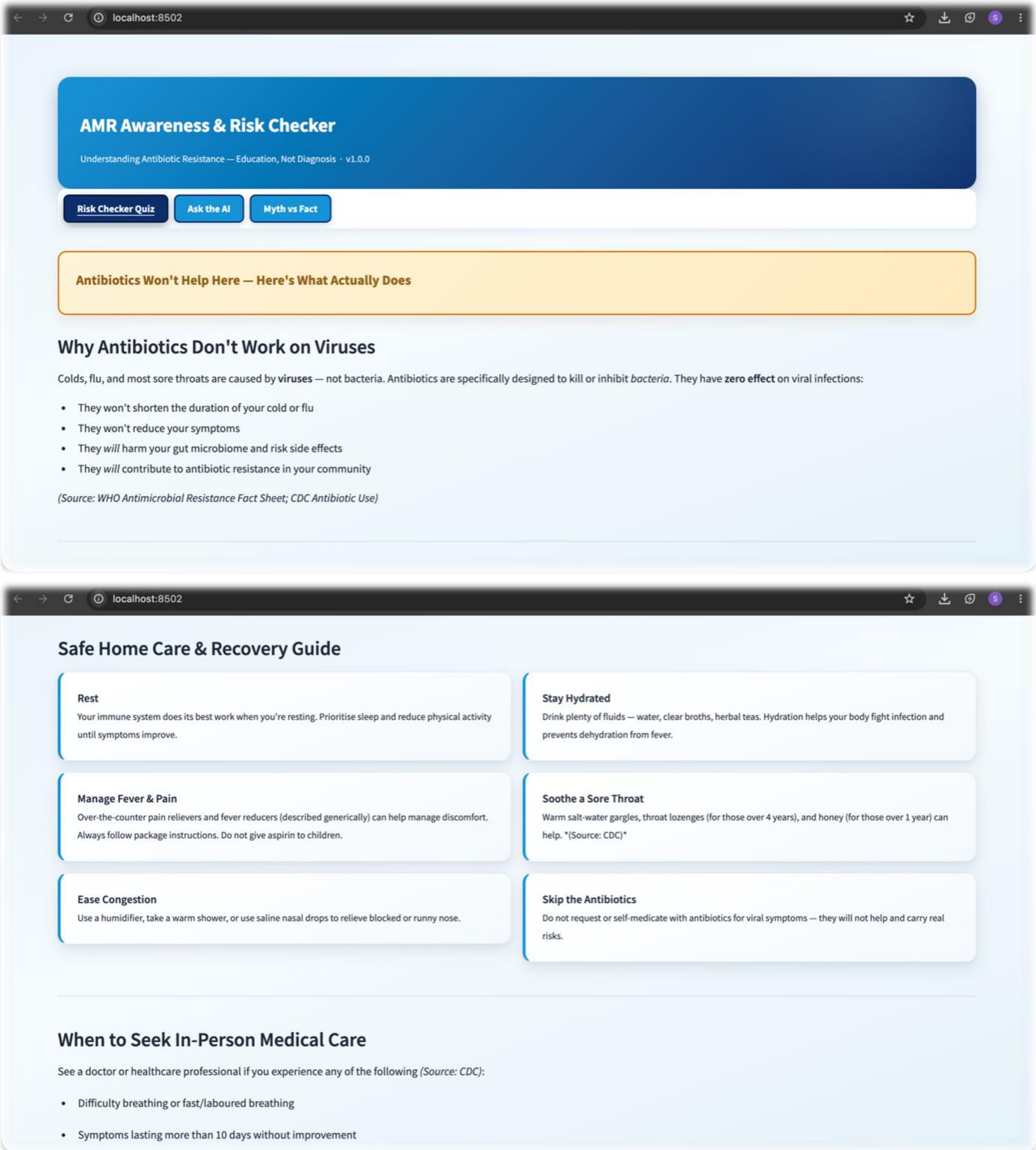
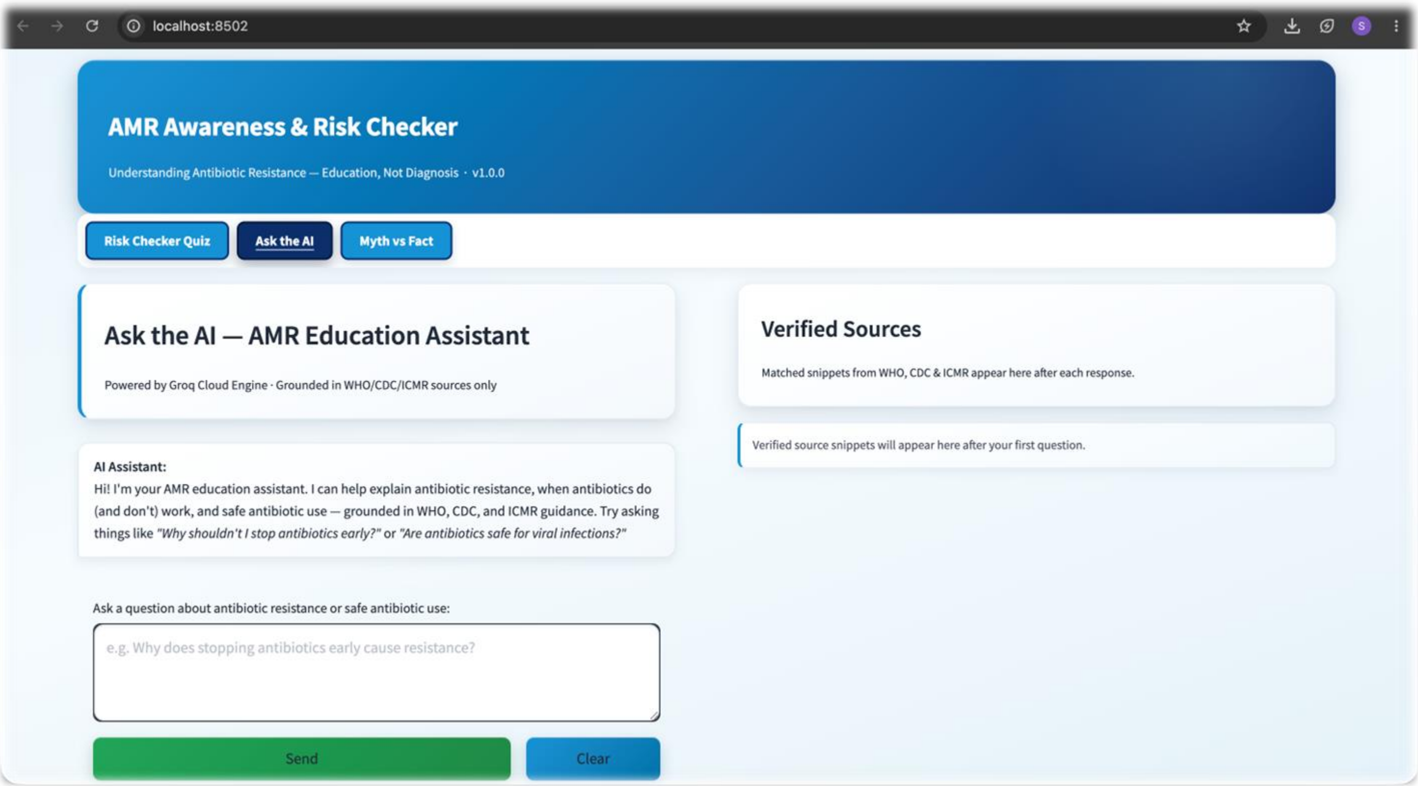


Fig4 & 5. Safe Home Care and Recovery Guide

The fourth interface presents the Safe Home Care & Recovery Guide. The system dynamically displays this page if a user's inputs point to a viral infection rather than a bacterial illness, actively steering them away from unnecessary antibiotic use.

Key features of this webpage include:

1. **Supportive Recovery Cards:** The page organizes self-care advice into distinct information cards covering rest, hydration, over-the-counter symptom management (fever, pain, sore throat, congestion), and explicit reminders to avoid unprescribed antibiotics.
2. **Red-Flag Escalation:** A dedicated lower section highlights critical scenarios requiring immediate professional medical attention, such as breathing difficulties, symptoms lasting over 10 days, or worsening clinical conditions.
3. **Antimicrobial Stewardship:** By balancing practical home care with clear boundaries for medical evaluation, the module discourages self-medication while providing safe, actionable health guidance.



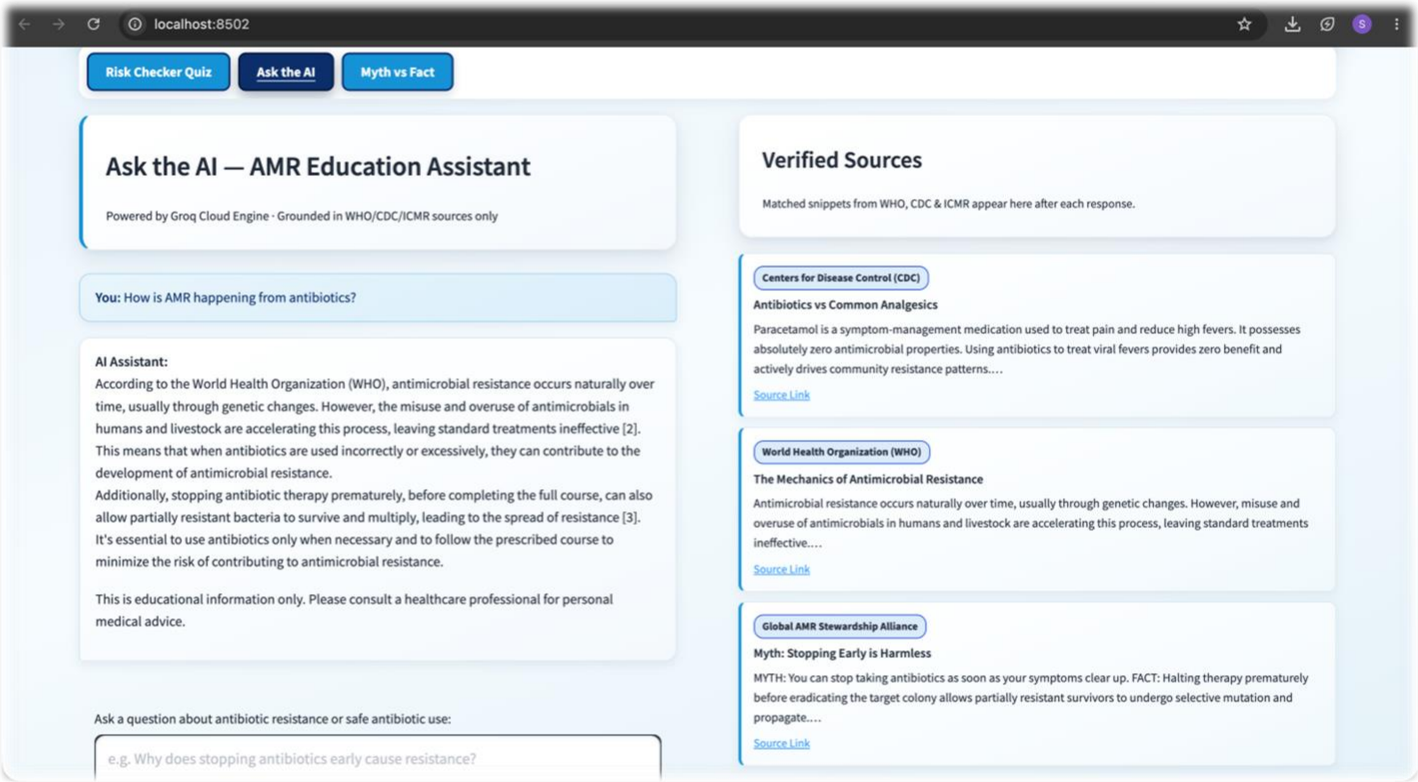
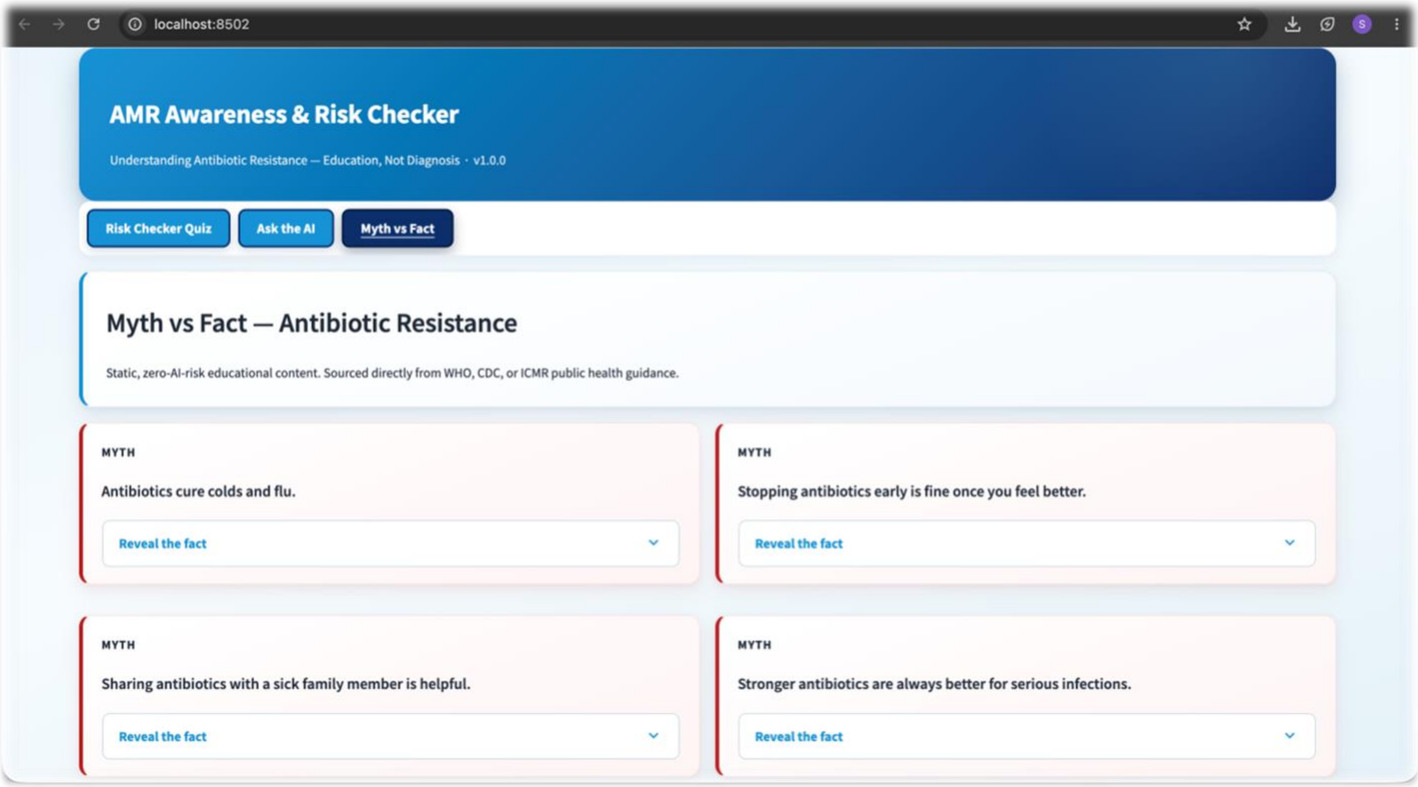


Fig6 & 7. AI-Powered AMR Education Assistant

The fifth interface showcases the "Ask the AI" module, built to deliver reliable education on antimicrobial resistance and safe antibiotic practices.

Key features of this webpage include:

1. **Grounded AI Responses:** Unlike open-ended chatbots, this assistant answers user queries about resistance, usage, misconceptions, and prevention using responses strictly grounded in WHO, CDC, and ICMR guidelines.
2. **Verified Sources Panel:** Alongside every AI response, a dedicated panel displays the exact evidence-based reference snippets used, giving users total transparency to check the facts themselves.
3. **Credible Information Delivery:** By tying large language model capabilities directly to official public health data, the page minimizes misinformation and provides accurate, context-aware public education.



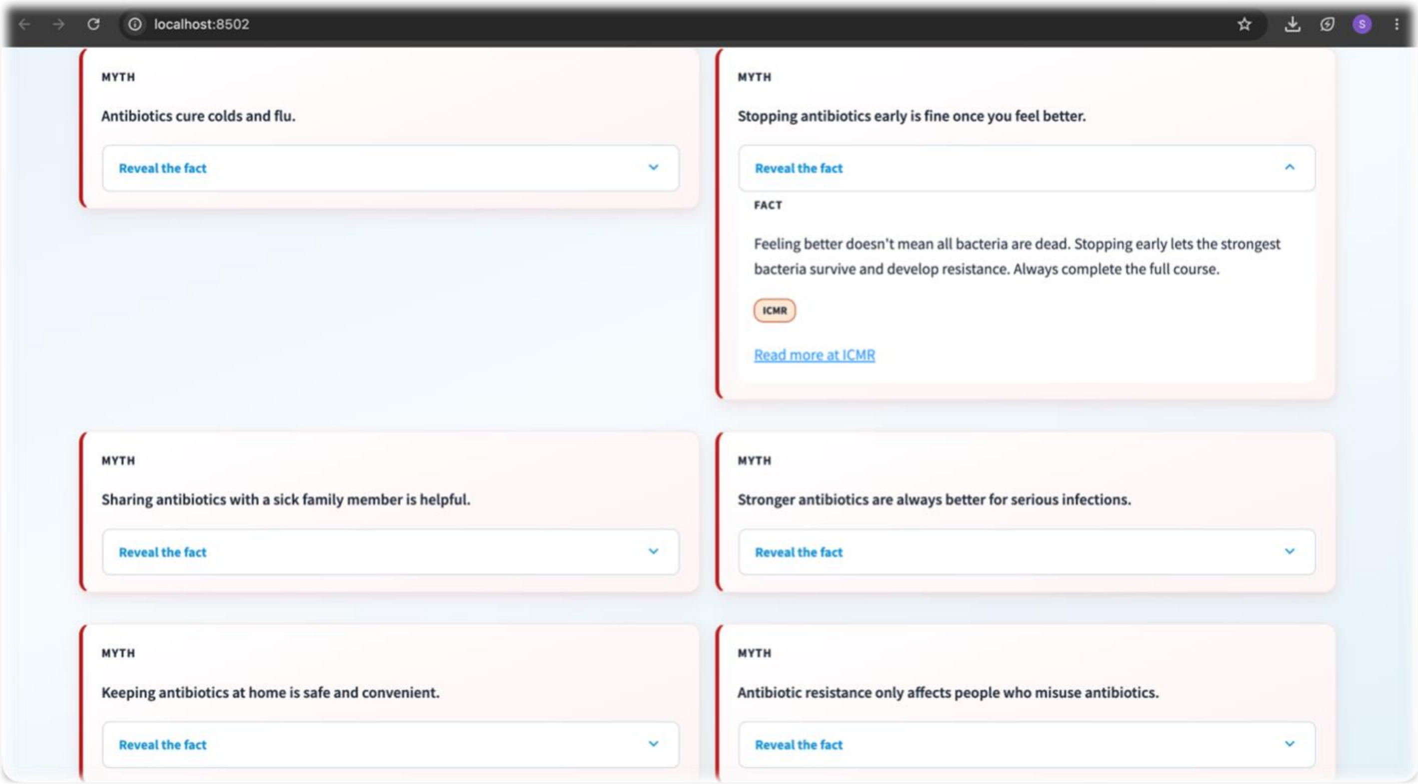


Fig8 & 9. Myth vs Fact – Antibiotics and Viral Infections

The sixth interface presents the "Myth vs. Fact" module, an educational section designed to dispel common misconceptions surrounding antibiotic misuse.

Key features of this webpage include:

1. **Evidence-Based Banners:** A prominent information banner clarifies that antibiotics target bacteria rather than viruses, providing immediate guidance against taking them for the common cold, flu, or sore throats.
2. **Consequence Tracking:** The page clearly lists the real-world impact of misuse, educating users on how unnecessary consumption disrupts the gut microbiome, triggers adverse drug reactions, and accelerates antimicrobial resistance.
3. **Global Guidelines:** All content is linked directly to official WHO and CDC guidance, ensuring the educational material is accurate, trustworthy, and validated.

8. Technology Stack

The core Technical Stack consists of the following tools and frameworks:

1. **Backend & Processing:** Python serves as the primary language due to its robust ecosystem for rapid development, data processing, and AI integration.
2. **Frontend Framework:** Streamlit provides an interactive web layout with minimal overhead, styled with custom HTML and CSS to deliver a modern, healthcare-inspired user interface.
3. **Conversational Intelligence:** The chatbot utilizes an OpenAI-compatible API endpoint connected to Groq-hosted language models, grounded strictly in public health data from the WHO, CDC, and ICMR.
4. **Security & Environment:** python-dotenv handles secure environment management, keeping sensitive API credentials entirely outside the source code.

9. Software Design Decisions

Key Software Engineering Decisions implemented during development include:

1. **Centralized Configuration:** Variables are consolidated into a single configuration module to streamline maintenance and bolster security.
2. **Deterministic Scoring:** The assessment relies entirely on predefined rules rather than AI-driven predictive modeling, guaranteeing transparent results.
3. **Custom UI Design:** Extensive custom CSS provides high accessibility, readability, and a unified appearance across all modules.
4. **Ethical AI Safeguards:** Mandatory disclaimers, constrained educational prompts, and strict chatbot restrictions align the system with responsible AI principles and healthcare ethics.

10. Module-wise Implementation

The project's Implementation Summary highlights a modular design built for high maintainability, fast debugging, and rapid hackathon development:

1. **Safety & Entry:** The application opens with a mandatory disclaimer establishing that the content is strictly educational and not medical advice, unlocking the dashboard upon acceptance.
2. **Risk Assessment:** Uses a structured questionnaire on antibiotic misuse paired with deterministic, rule-based scoring—ensuring full transparency and reproducible results without relying on machine learning.
3. **Educational AI Assistant:** Answers user questions regarding resistance and prevention within strict educational limits. It displays verified, authoritative source snippets right next to its answers to build user confidence.
4. **Myth vs. Fact:** Reinforces proper health practices through concise, evidence-based statements from trusted healthcare bodies, providing a comprehensive awareness platform that never replaces professional care.

11. User Interface and Experience

The project's User Interface and Experience (UI/UX) focuses on a clean, accessible layout for diverse users:

1. **Visual Identity:** A healthcare-inspired blue color palette establishes trust and professionalism.
2. **Layout & Style:** Custom CSS powers cards, badges, expandable sections, and responsive elements to optimize navigation and readability.
3. **Accessibility:** Designed with high text contrast and minimal visual clutter, presenting complex educational facts in small, digestible segments.
4. **Streamlit Customization:** Standard Streamlit widgets were tailored to keep a highly consistent aesthetic across interactive features like the questionnaire, chatbot, and Myth vs. Fact section.

12. Challenges Faced During Development

The project's Technical Challenges and Solutions summarize the main development hurdles:

1. **Credential Management:** Securely managing API tokens and environment variables was resolved by implementing environment files and a centralized configuration module.
2. **Widget Bias:** Default questionnaire selections risked biasing user answers. This was fixed by configuring Streamlit widgets to require explicit, active user input.
3. **CSS & Rendering Conflicts:** Extensive custom styling caused layout clashes with Streamlit's default theme. We iteratively debugged HTML rendering, Markdown interpretation, and text visibility to ensure visual consistency.
4. **Chatbot Guardrails:** Preventing the AI from generating diagnostic or treatment advice required tight prompt engineering. The assistant was strictly restricted to providing educational material grounded in verified public health resources.

12.1. Lessons Learned

The project's Key Learnings summarize the takeaways from the development process:

1. **Core Development Pillars:** The process underscored the vital role of modular software design, responsible AI deployment, secure configuration management, and user-centered interface design.
2. **Hybrid Architecture Value:** Pairing deterministic rule-based logic with a strictly constrained conversational AI proved highly effective for balancing interactive engagement with user safety.
3. **Iterative Refinement:** The experience reinforced that continuous debugging, testing, and structural refinement are essential when building trustworthy educational tools for the healthcare domain.

13. Results and Evaluation

The application was rigorously evaluated against its original design goals, demonstrating excellent performance in safety, usability, and architecture:

1. **Objective Met:** Successfully delivers a structured educational journey—spanning safety disclaimers, risk assessments, and interactive myth-busting—to promote responsible antibiotic practices.
2. **Hybrid Design Success:** Combines deterministic scoring with tightly constrained conversational AI to maximize result transparency and completely eliminate medical hallucinations.
3. **Enhanced Accessibility:** Leverages custom-styled cards, badges, and expandable UI workflows to seamlessly translate complex health guidelines into digestible public insights.
4. **Proven Stability:** Rigorous testing and iterative debugging verified full application reliability, fixing rendering issues and questionnaire biases to ensure a dependable user experience.

14. Social Impact

This application addresses the global challenge of antimicrobial resistance by shifting focus from clinical diagnosis to interactive public education:

1. **Public Health Education:** Contributes directly to public awareness of antimicrobial resistance rather than clinical diagnosis.
2. **Behavioral Change:** Encourages finishing prescribed courses, avoiding self-medication, distinguishing bacterial from viral infections, and seeking professional consults.
3. **Responsible AI Deployment:** Demonstrates how artificial intelligence can safely complement existing health initiatives without offering unsupported medical advice.
4. **Community Outreach:** Delivers trusted information in an engaging format designed to raise AMR awareness among students, families, and the general public.

15. Future Scope

While the application successfully meets its current goals, several opportunities exist to expand its scope and reach:

1. **Expanded Accessibility:** Incorporating multilingual support to engage different regions of India, alongside voice interactions and mobile application deployment.
2. **Enriched Content:** Integrating additional verified healthcare datasets to safely expand the AI assistant's knowledge base.
3. **Interactive Engagement:** Developing gamified learning tracks, periodic awareness quizzes, and targeted, institution-specific educational campaigns.
4. **Impact Tracking:** Utilizing anonymous analytics dashboards to measure user engagement and evaluate the overall effectiveness of the public health intervention.

16. Conclusion

The AMR Awareness & Risk Checker successfully demonstrates that responsible artificial intelligence, deterministic software engineering, and evidence-based healthcare guidance can be seamlessly integrated into a single educational platform. Throughout development, priority was strictly placed on transparency, safety, and public awareness over automated diagnosis or medical treatment. Beyond providing valuable, hands-on experience in engineering and safe AI integration, this project underscores the massive potential of interactive digital tools to strengthen antimicrobial stewardship. Ultimately, the completed system serves as a meaningful contribution to public health education and forms a trusted foundation for future medical awareness applications.

References

1. **World Health Organization (WHO):** Antimicrobial Resistance Fact Sheets and Public Health Guidance
2. **Centers for Disease Control and Prevention (CDC):** Antibiotic Use and Antimicrobial Stewardship Resources
3. **Indian Council of Medical Research (ICMR):** National AMR Guidelines and Educational Resources
4. **Streamlit Documentation.**
5. **Python Documentation.**
6. **Groq API Documentation.**
7. **Google Gemini / Claude / ChatGPT**

****Honest AI Declaration****

In the spirit of technical transparency, AI was utilized selectively during the project lifecycle to accelerate the initial drafting of the Product Requirement Document (PRD) and generate foundational, elementary code templates.

Manual engineering and intervention were strictly relied upon to observe and correct user interface problems, refine the design for optimal usability, and ensure continuous adherence to the project's fundamental core requirement: providing educational insight without crossing into diagnostic interaction.